Harsh Raj

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EDUCATION

Jain (Deemed-to-be University), Global Campus Bengaluru, India
B.Tech in Computer Science and Engineering (AI & ML) *Aug 2025 – Jun 2029*
CGPA: 8.97 / 10 | **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming (Java), Database Management Systems, Discrete Mathematics, Linear Algebra, Probability and Statistics

TECHNICAL SKILLS

Languages: Python, Java, C, HTML, CSS
Web and APIs: React, FastAPI, REST APIs, Tailwind CSS, Vite
Machine Learning and GIS: scikit-learn (Random Forest), OSMnx, Leaflet, OpenStreetMap
Developer Tools: Git, GitHub, Vercel, VS Code

PROJECTS

LandSafe | *Python, FastAPI, scikit-learn, OSMnx, React, Leaflet* GitHub

- Built a landslide early-warning and evacuation system for Smart India Hackathon 2026 (team shortlisted for Round 2), covering India with a focus on the North Eastern states.
- Trained a Random Forest model that predicts landslide risk from rainfall and slope with **91% accuracy**.
- Designed a three-stage Predict, Route and Verify pipeline that computes evacuation routes on OpenStreetMap road networks using OSMnx.
- Served the pipeline through **6 FastAPI REST endpoints**, consumed by a React and Leaflet dashboard with a live GIS hazard map.

NammaRoute | *React, Tailwind CSS, FastAPI, OpenStreetMap* GitHub

- Building a multimodal trip planner for Bengaluru; the working prototype plans routes between any two places, combining bus, metro, walking and auto.
- Built search over streets, campuses and landmarks in Bengaluru from an OpenStreetMap place index.
- Integrated OpenRouteService for road geometry, keeping the API key on the server.

Personal Portfolio Website | *React, Vite, Tailwind CSS, GSAP, Three.js, Vercel* Live Site

- Built and deployed a responsive portfolio with a 3D photography gallery and a working contact form.
- Cut the gallery's image payload by **95%** (33.6 MB to 1.8 MB) by converting photos to resized WebP and lazy-loading them.
- Removed per-frame React re-renders by driving animations through refs and requestAnimationFrame.

Java and Coursework Projects | *Java, Swing, OOP*

- Built three Swing games (Snake, Chess, Tower Defense) of increasing complexity, applying inheritance, interfaces and multithreading.
- Built a Student Grade Management System in Java and rebuilt it as a web app on GitHub Pages.
- Built physics web apps for coursework: a Bloch sphere qubit simulator and a LiDAR distance scanner.

ACHIEVEMENTS AND CERTIFICATIONS

- **Smart India Hackathon 2026:** Cleared Round 1, the university internal hackathon (Sep 2026), with LandSafe; team shortlisted for Round 2 (in progress).
- **LinkedIn Learning (Sep 2026):** Programming Foundations: Discrete Mathematics; Learning Graph Neural Networks.

LEADERSHIP AND ACTIVITIES

Turing Club, Jain University Global Campus Bengaluru, India
Photography Team Member *2026 – Present*

- Photograph and film college fests, club events and tech events, and edit event videos.